

INTRODUCTION TO ECONOMICS

MGT404

COURSE REVIEW

Yale

First... Course Evaluation Please 😊

Exam Details

FINAL EXAM

Fall 2024

- This is a closed-book exam. Only a 8.5"x11" formula sheet is allowed.
- No electronic devices are allowed, except for the simple calculator we provide.
- You may not communicate with other students during the exam. You will also keep the exam contents confidential.
- There are **180 points in the exam**. You also have **180 minutes**.
- Do not be concerned if your answers are not whole numbers. Large numbers are avoided to lower the burden of calculation. Don't worry if you find the numbers unrealistic---you could make sense of these numbers in your head by picking the right units yourself (but no need to answer with these units).
- Please be as complete as possible. Write down the reasonings for your answer (unless instructed otherwise).
- Complete all **SHORT ANSWER** questions in one Blue Book, and all **LONG PROBLEMS** in a 2nd Blue Book. **Write your name and "SHORT" / "LONG" on the cover of your Blue Books.**
- Answers written **anywhere else, including on the question sheet** will **NOT** be graded.
- Please return the **Blue Books** AND **the question sheet** after the exam.
- A camera will be set in the classroom to monitor exam situations (Exam recordings will be deleted once final grades are posted unless required for an ongoing Honor Committee case).
- Faculty will circulate in each classroom throughout the exam to answer clarifying questions.

Review

Consumers

Reservation Prices

Demand Curve

A demand curve represents the distribution of *reservation prices* in a market.

Always add demand curves in terms of **quantity**.

Inputs: Marginal Productivity, Cost

Optimal Production Mix

Supply Curve

A firm's marginal cost curve (based on its optimal production mix) is its supply curve.

Market supply is the sum of firm supply (add by **quantity**).

Market Equilibrium

Quantity Supplied = Quantity Demanded. No consumer wants to buy more or less; no firm wants to supply more or less. This determines the market-clearing price and quantity, consumer and producer surplus, and deadweight loss.

Efficient (total surplus) and Predictive!

Summary of Elasticity and Incidence

Own-Price Elasticity

- If I increase price, how much will quantity respond?
 - A lot (more than proportional): *elastic*
 - Not a lot (less than proportional) *inelastic*
- Note: always negative. Increasing my price means *decreasing* my quantity sold.

$$\varepsilon = \frac{\partial Q}{\partial P} \cdot \frac{P}{Q} = \frac{\Delta Q}{Q_1} / \frac{\Delta P}{P_1}$$

Cross-Price Elasticity

- If another good increases its price, how will my quantity respond? No longer only negative!
 - Positive: substitutes.
 - Negative: complements.
- Managerial implication: who wins when someone else raises prices?

Income Elasticity

- If my income goes up, do I buy more or less?
 - More: normal or luxury goods
 - Less: inferior goods.

Drawing supply/demand curves: elastic is flat, inelastic is steep.

Own-Price Elasticity

$$\varepsilon = \frac{\% \text{change in } Q}{\% \text{ change in } P} = \frac{\frac{Q_2 - Q_1}{Q_1}}{\frac{p_2 - p_1}{p_1}} = \frac{\partial Q / Q}{\partial P / P} = \frac{\partial Q}{\partial P} \cdot \frac{P}{Q}$$

If I increase my price by 1%....

Quantity falls by less than 1%

- My demand is “inelastic”. It doesn’t respond much to price.

Quantity falls by more than 1%

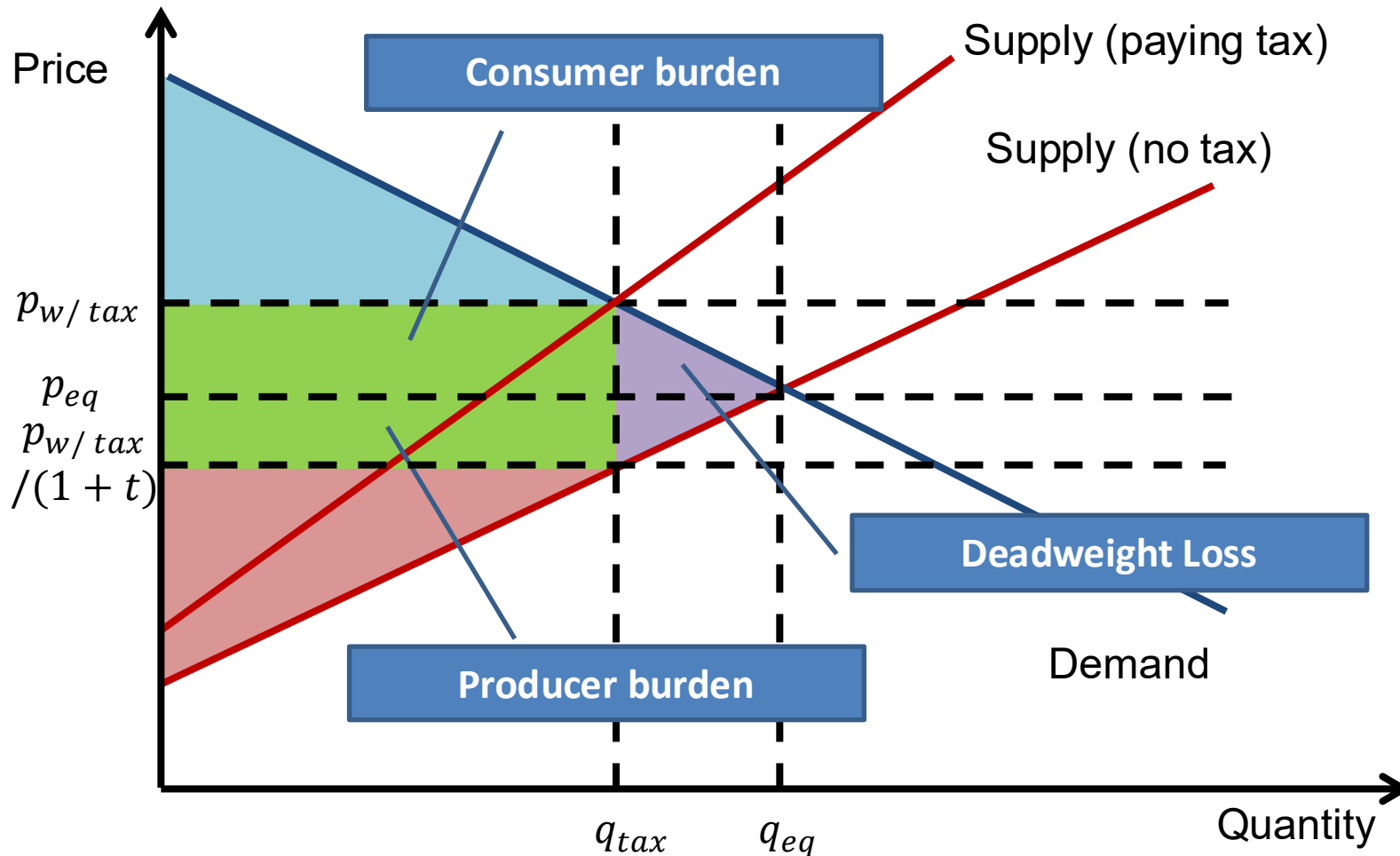
- My demand is “elastic”. It is very responsive to price.



If I am a firm with market power, and my demand is “inelastic” – I should raise price! I can increase price, and lose few customers, and selling fewer units is more profitable.

“Incidence” of Taxes

Suppose there is a percentage tax imposed on manufacturers:



Solving for a Market Equilibrium

A group of local activists in Cambridge, Massachusetts is trying to encourage lawmakers to expand the city's use of taxes to reduce consumption of products with negative externalities. Specifically, the group believes that the city should tax beer at restaurants, reasoning that higher beer prices would mean fewer drunk college students, who (they argue) make a lot of noise and upset downtown residents. The group favors imposing a tax of \$2 per beer served at restaurants. For this question assume there is no baseline alcohol tax.

Cambridge restaurants provide beer according to the following weekly supply curve:

$$P_S(Q) = 2 + \frac{1}{5}Q,$$

and face the following weekly demand curve:

$$P_D(Q) = 12 - \frac{1}{5}Q,$$

where Q is in 1,000s of bottles per week.

- a) (4 points) What is the market equilibrium price of beer sold in a restaurant in Cambridge and the total quantity sold per week with no tax?

Price = _____

Quantity = _____

What are the steps to solving?

Solution

We can solve this by first equating the prices and then solving for the equilibrium quantity:

$$P_s(Q) = P_d(Q)$$

$$2 + \frac{1}{5}Q = 12 - \frac{1}{5}Q$$

$$\frac{2}{5}Q = 10$$

$$Q^* = 25$$

Given the units, this means that 25,000 bottles of beer are sold each week in Cambridge. Evaluating the market demand at $Q^* = 25$ gives us the equilibrium price:

$$P_d(25) = 12 - \frac{1}{5}(25)$$

$$= 12 - 5 = 7$$

So the equilibrium price is \$7 per bottle.

How does a tax change this?

- Per unit tax: shift supply curve by tax amount
- Percent tax: multiply supply curve by $(1+\text{tax})$.

b) What would the new market equilibrium be if the beer tax were in place?

Price (inclusive of tax): _____

Quantity: _____

c) What is the total tax revenue? What percent of the tax is borne by consumers?

Total tax revenue) (\$): _____

% paid by consumers (\$): _____

Solving for a VC curve...

1. Start with a production function and prices:

$$q = f(x_1^*, x_2^*) \quad p_{x_1}, p_{x_2}$$


2. Equate the return from investing in inputs. This determines the optimal ratio of inputs to produce at lowest cost.

$$MPI_{x_1} = MPI_{x_2}$$
$$\frac{\frac{\partial f}{\partial x_1}(x_1, x_2)}{p_{x_1}} = \frac{\frac{\partial f}{\partial x_2}(x_1, x_2)}{p_{x_2}}$$


**MATH BELOW NOT
TESTED**

3. Now you'll have a relationship between x_1^* and x_2^* . Rewrite $q = f(x_1^*, x_2^*)$ as a function of just one variable, e.g. $q = f(x_1^*)$
Rearrange to solve for the quantity of input as a function of output, e.g. $x_1^*(q)$
Use the relationship between x_1^* and x_2^* to also get $x_2^*(q)$
- 

Now, variable cost is just the inputs (as a function of q) times their prices:

$$VC(q) = p_{x_1} \cdot x_1^*(q) + p_{x_2} \cdot x_2^*(q)$$

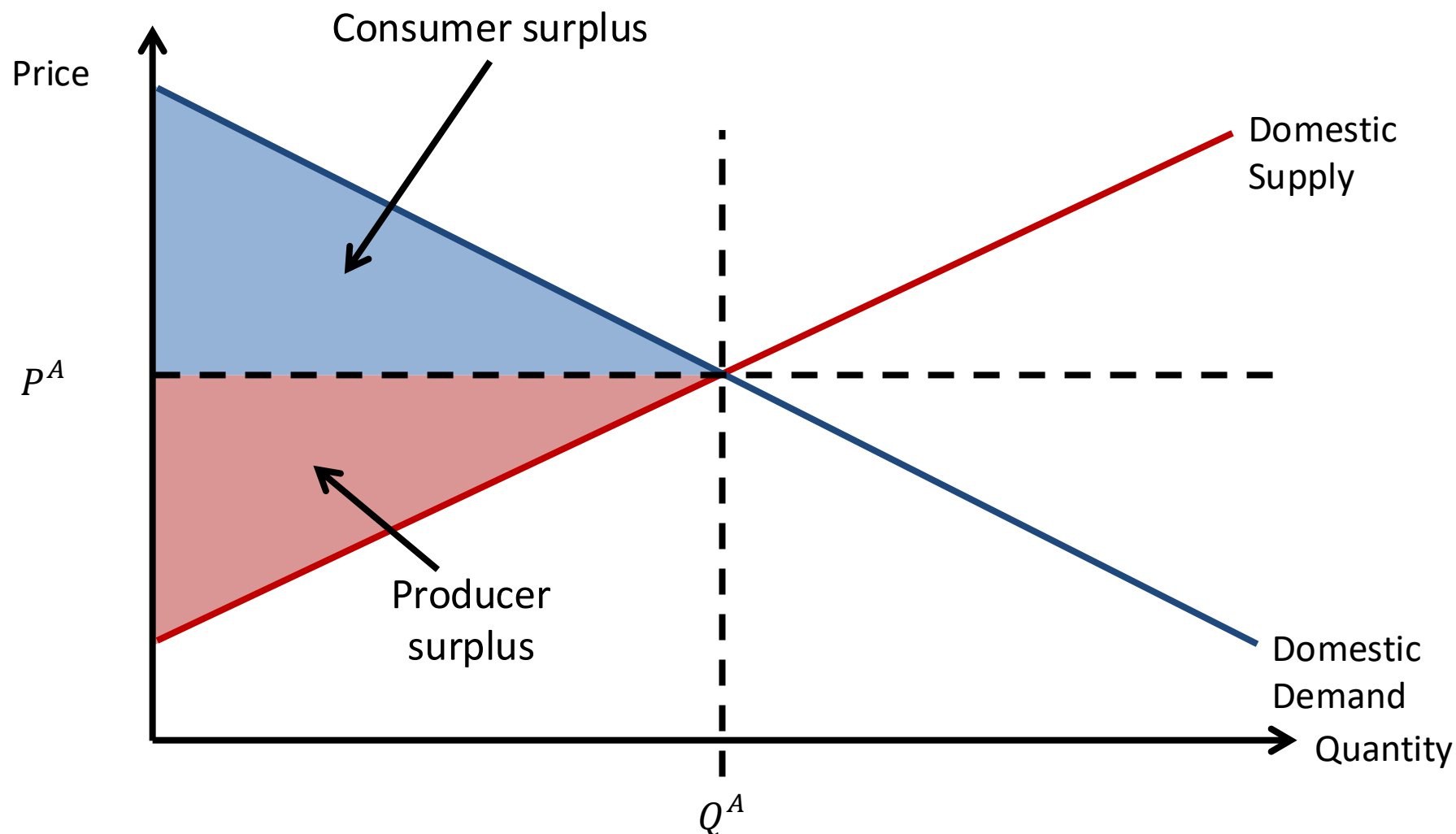
Practice!

Suppose you are given a production function:

$$q = f(l, m) = 6l + 4m + 2lm - 0.1lm^2$$

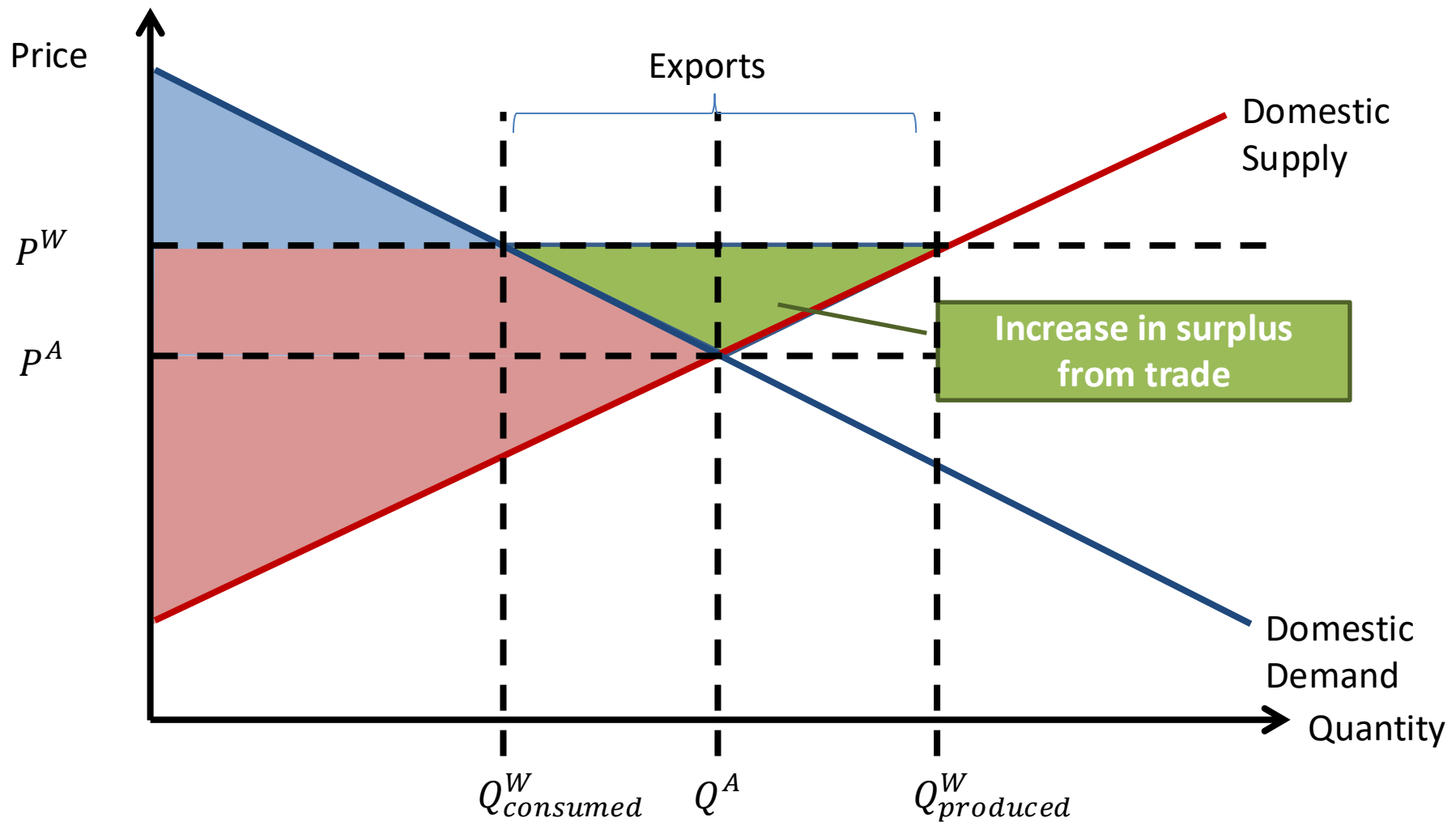
What are the marginal products of labor and capital?

Consider Autarky: Our Standard Market Analysis



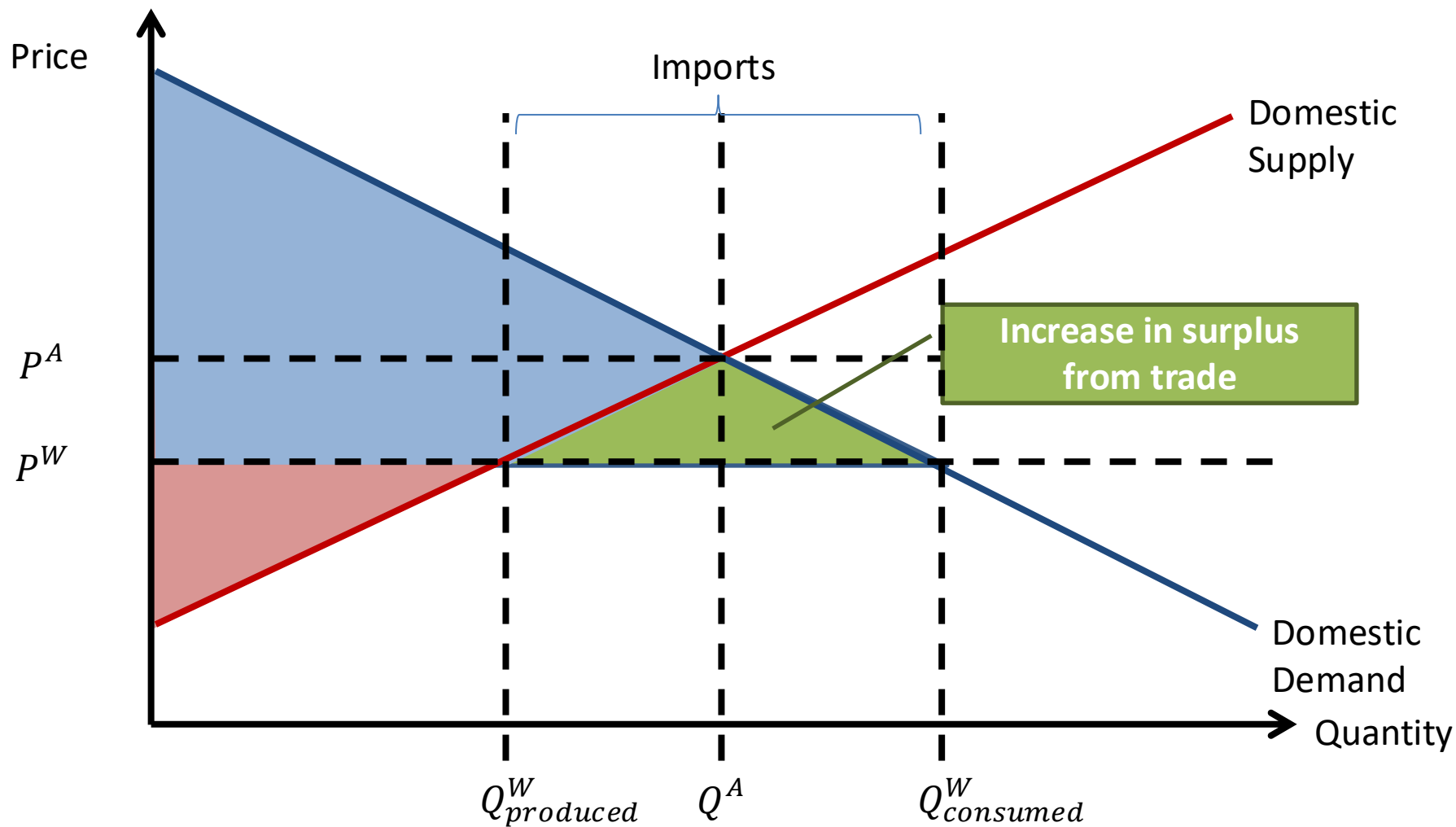
Autarky equilibrium: domestic supply = domestic demand

Free Trade Case 1: $P^W > P^A$:



Total domestic surplus increases, even though domestic consumer surplus falls.

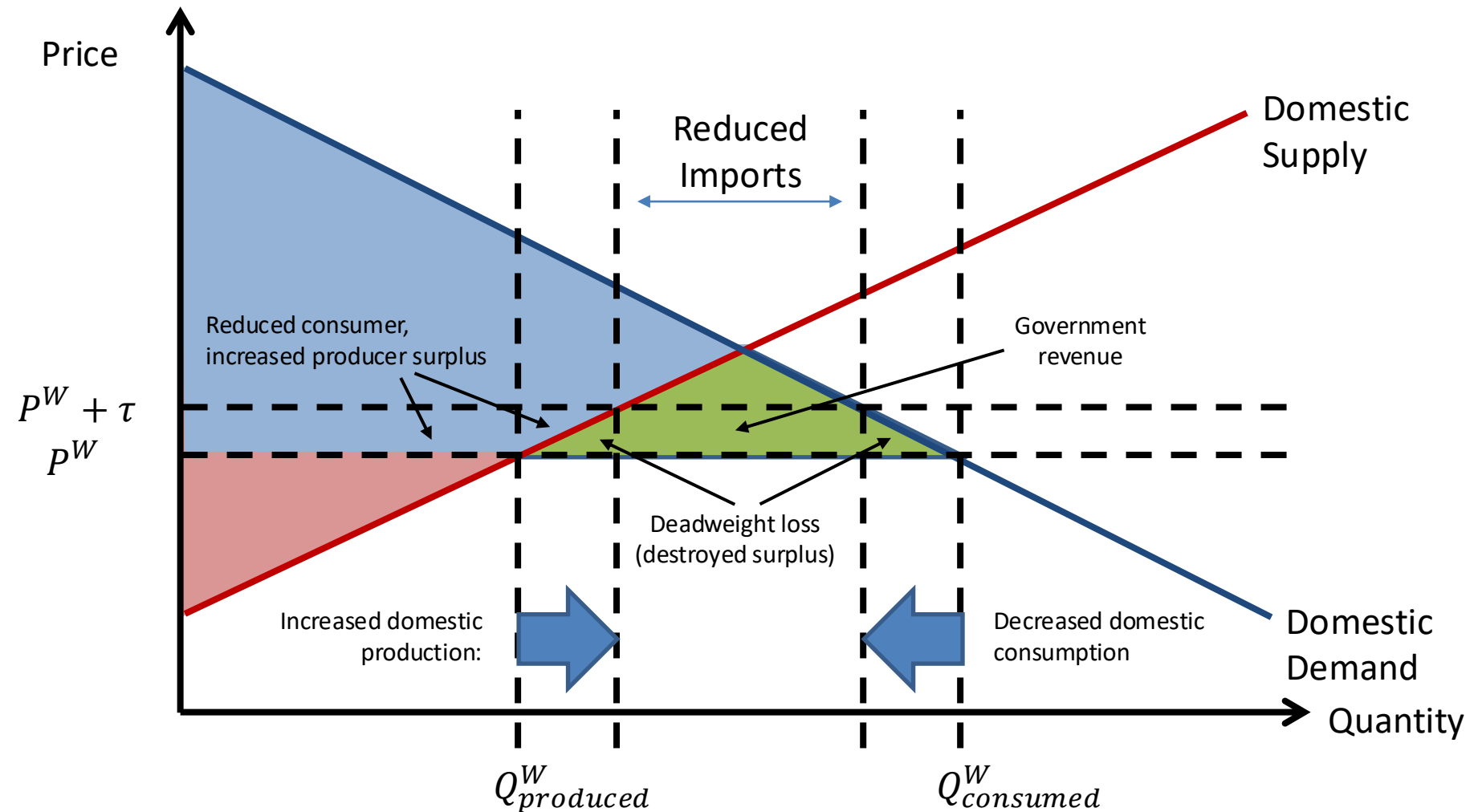
Free Trade Case 2: $P^W < P^A$:



Total domestic surplus increases, even though domestic producer surplus falls.

Import Tariff

An import tariff is a “tax”, τ , on the world price of a good, P^W .



Perfectly Competitive Firm Behavior

Firms Take Market Price as Given

Firms in perfectly competitive markets choose an optimal output quantity and whether or not to shut down (short run) or exit (long run).

Optimal quantity sets $MR(Q) = MC(Q)$, except $MR(Q) = P$ in this setting.

Market Price

Outcome

$$P > \min(ATC)$$

- Produce and earn a profit
 - Market price is above average cost

$$\min(AVC) < P < \min(ATC)$$

- Produce and earn a loss in the short run
 - Still earning variable profit, produce to cover fixed costs

$$P < \min(AVC)$$

- Shut down production
 - Selling any quantity would increase losses

In the Long Run, Firms May Exit

Firms will exit if $P < \min(ATC)$

Pricing with Market Power

Firms maximize profits

Firms face downward sloping demand. Underlying tradeoff:

- If I want to produce more, the market-clearing price will be lower (on all units).

Maximizing profits means:

- Derivative of profit function is zero. (I am at the point where I cannot increase profits by increasing output)
- Marginal Revenue equals Marginal Cost. (Same logic)

So, optimal quantity sets $MR(Q) = MC(Q)$.

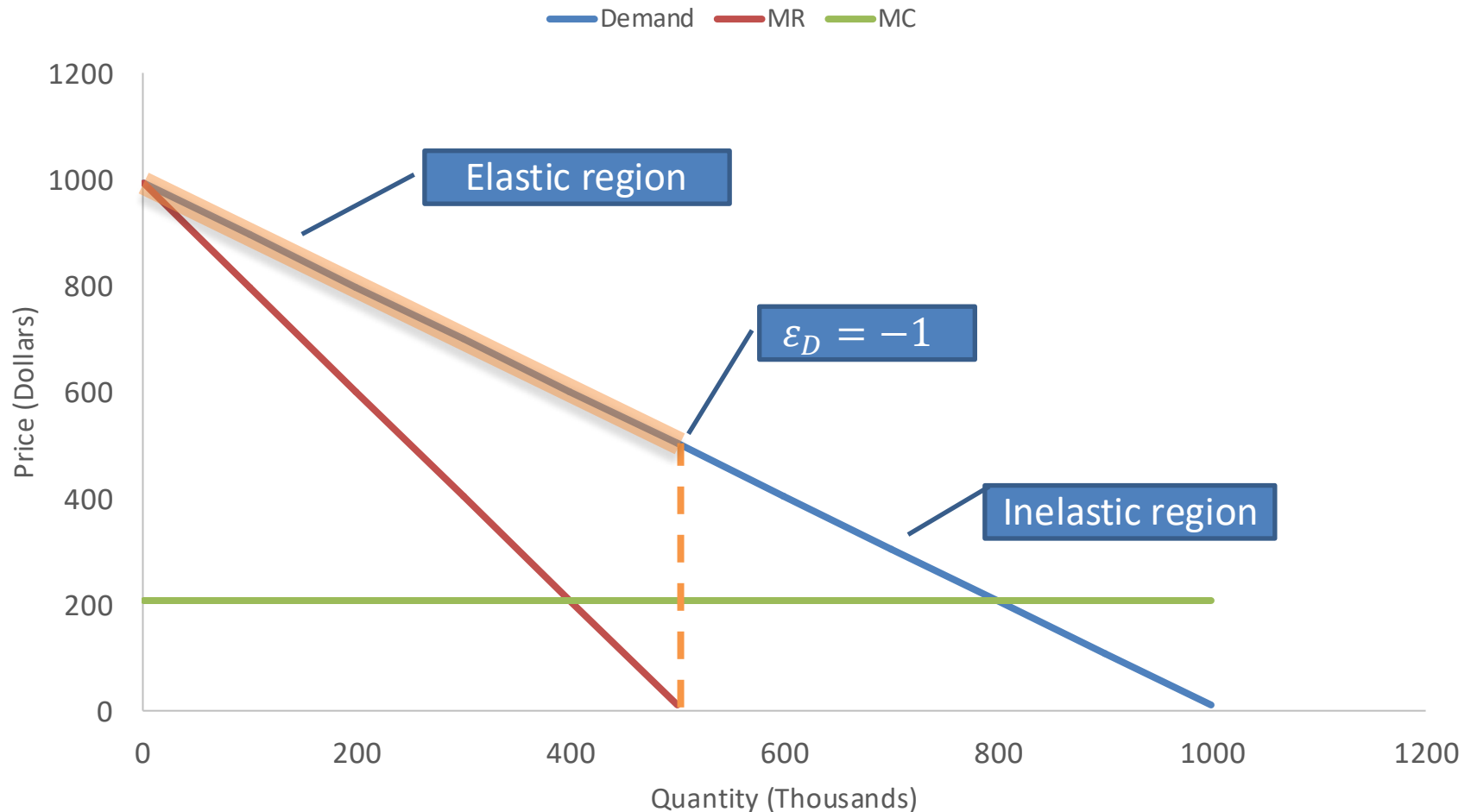
Write out the same ideas in terms of elasticity. Can express MR in terms of demand elasticity. This gets us to the Inverse Elasticity Pricing Rule (IEPR).

- Note: The IEPR equation holds if I am pricing optimally. If I am not pricing optimally, it does not hold.

Equivalently: the IEPR holds when $MR=MC$.

$$\frac{P - MC}{P} = \frac{-1}{\varepsilon}$$

Pricing with Market Power



Question: Why don't we price in the inelastic range?

If you can raise price without losing many customers, you aren't charging enough. Also, at $\epsilon_D = -1$, $MR=0$. Firms price "before" this point.

How to do MR=MC

Inputs

- Demand curve: $P(q) = 100 - 2q$
- Cost function: $C(q) = 25q$

Process

- First need revenue; take derivative to get marginal revenue.
- Derivative of cost is marginal cost. (Or you are given MC)

$$\begin{aligned}\text{Revenue} &= q \cdot P(q) = 100q - 2q^2 \\ \text{Marginal revenue} &= 100 - 4q\end{aligned}$$

$$\begin{aligned}\text{Marginal Cost} &= 25 \\ &(\text{derivative of } C(q))\end{aligned}$$

$$\begin{aligned}MR &= MC \\ 100 - 4q &= 25 \\ 75 &= 4q \\ 18.75 &= q^*\end{aligned}$$



$$\begin{aligned}P(q^*) &= 100 - 2(18.75) \\ &= 62.50\end{aligned}$$

Uniform Pricing

Pricing at the Taj Mahal. The Taj Mahal in Agra is the most popular monument in India, attracting thousands of visitors daily. The total daily demand for visits from foreign tourists is:

$$Q^F(p) = 4,000 - 200p,$$

where p is measured in US dollars. The Taj Mahal's operator, the Agra Development Authority (ADA), incurs a cost of \$0.50 for every visitor who tours the Taj Mahal.

(a) (4 points). Find the profit-maximizing price that the ADA should charge to foreign tourists. How much profit does the ADA make every day?

Optimal price: _____

Daily profit: _____

Solution:

$$Q^F(p) = 4,000 - 200p \rightarrow P = 20 - 0.005Q$$

$$\text{Profits} = \text{Revenue} - \text{Cost} \\ = P(Q) \cdot Q - TC(Q)$$

Solution 1: Since cost is constant, write profits as: $\pi = (20 - 0.005Q)Q - 0.5Q$

$$\frac{d\pi}{dQ} = 20 - 0.01Q - 0.5 = 0 \rightarrow Q = 1950$$

Derivative to find a maximum

$$P(1950) = 10.25$$

$$\pi = 19,012.50$$

Solve for a price at that quantity.

Evaluate profit function at $Q=1950$

Solution 2: Set $MR=MC$

Revenue Function

$$\begin{aligned} R &= P(Q) \cdot Q \\ R &= (20 - 0.005Q) \cdot Q \\ R &= 20Q - 0.005Q^2 \\ MR &= 20 - 0.01Q \end{aligned}$$

Marginal Revenue: Derivate of Revenue Function

Marginal Cost, from question

$$\begin{aligned} MC &= 0.5 \\ MR &= MC \\ 20 - 0.01Q &= 0.5 \\ 19.5 &= 0.01Q \\ Q &= 1,950 \\ P(1950) &= 10.25 \end{aligned}$$

Use rule that $MR=MC$ at optimal output

Sub in and solve for Q

Price at optimal quantity

$$\pi = 19,012.50$$

Pricing with Market Power Topics

Topic (Lecture)

Setting

Approach & Logic

Uniform Pricing (8)

Math Example:

- Oculus Rift Pricing

- You face downward sloping demand
- You set a single per-unit price for your good.

- A firm wants to maximize profits.
- Maximize $\pi(Q)$, set $MR(Q) = MC(Q)$, or use the IEPR: $\frac{P^* - MC}{P^*} = \frac{-1}{\epsilon}$
- Recall: $R(Q) = P(Q) \cdot Q$
- Less elastic demand = higher markup.

Perfect Price Discrimination (9)

Math Example:

- Safeway Bakery

- You know each consumer's reservation price for each unit
- Resale is not possible

- Charge each consumer their exact reservation price for each unit
- Firm captures all surplus in the market.

Demographic Price Discrimination (9)

Math Example:

- Amtrak Student vs Business

- There are different types of consumers
- You can verify which type a consumer is (e.g. student)
- Set optimal uniform prices for each type of consumer.
- This “separates” the market into a market for each consumer type.
- Compare to pricing to the aggregate demand curve.

Pricing with Market Power Topics

Topic (Lecture #)

Self-Selecting Pricing (9)

Math Example:

- Golf Course

Setting

- Two types of consumers want a single good
- You cannot verify the types, sell “low” and “high” goods.
- “Willing to buy”: low types will buy low good.
- “Buy the right one”: high types buy high good.

Approach & Logic

Choose highest profit among:

1. Only high types: charge high type reservation price for high quality good
2. Both types, one price: charge highest low-type reservation price
3. “Menu”: low-quality good at low-type reservation price; choose high-q price so that high types just willing to pay

Quantity Discount (Single Type) (10)

Math Example:

- Gym Membership

- Consumers have demand for multiple units of the good.
- You can charge an upfront fee (membership)
- Set the “membership fee” to be the total consumer surplus.

Self-Selecting Price Discrimination

We will now set prices for a “High type” good and a “Low type” good

The “**Willing to Buy**” Constraint

- Each type must get non-negative consumer surplus, so they’ll purchase at all.
- Most important for “low types”.

The “**Buy the Right One**” Constraint

- You are concerned the “high types” will get more surplus from the “low type” product, since it is cheap.
- So you need to “leave some money on the table” (consumer surplus) so they’re happy about choosing the higher priced good.

Binding constraints:

Will the low type be willing to buy the low type good?

“Low type value of low good” ≥ 0

Will the high type prefer the high good to the low good?

“High value of high good” $\geq$
“High value of low good”

Pricing with Market Power Topics

Topic (Lecture #)

Menu Pricing (10)

Math Example:

- Audible. Excel sheet.

Setting

- Two types of consumers have different demands for multiple units.
- Cannot verify types.
- Cannot allow any quantity.

Approach & Logic

Pick optimal size of “small” package:

- Large size comes from where high type demand meets marginal cost. Large price leaves high types enough surplus so that they are willing to buy.
- Price of small is the low-type consumer surplus from small.
- Pick small size to maximize profits.

Two-Part Tariff (10)

Math Example:

- Zipcar. Excel sheet.

- Same as above, except can now allow any quantity.

To serve both types:

1. Set membership fee to low-type consumer surplus for a usage price p
 2. Write out profits from membership and usage as a function of p
 3. Find the p that maximizes profits.
- Check that it is worth serving both types.

Risk & Insurance

When choosing between options, agents use **expected utility**:

Expected Value

$$EV = p_1x_1 + p_2x_2 + \dots$$

Implies *risk-neutral*...

Expected Utility

$$EU = p_1u(x_1) + p_2u(x_2) + \dots$$

Allows for *risk-averse*.

Utility Functions that allow risk aversion:

Most common are $u(x) = \sqrt{x}$.

Two important concepts in risk:

Certainty Equivalent

The price at which you would be willing to sell a risky prospect (which could be negative).

- Solve for c where $p_1u(x_1) + p_2u(x_2) + \dots = u(c)$

Risk Premium

The difference between the expected value and the certainty equivalent of a risky prospect.

Actuarially Fair Insurance (if you're risk-averse, you would fully insure at this price)

If insurance is priced to break even at expected value, it is actuarially fair.

Example:

Try to solve the following problem:

Situation

- Your utility is $u(x) = \sqrt{x}$
- You have zero savings, and face a risk of receiving \$100 (25%), \$200 (50%), and \$500 (25%) this week.

EV

- What is the expected value of this risk?

Certainty Equivalent

- What is the guaranteed outcome $\$c$ you would be willing to trade for this risk?

$$EU(risk) = EU(certain)$$

$$EU(risk) = \sqrt{\$c}$$

(Risk Premium): How much EV would you give up to get rid of the risk?

CE is asking: What is my reservation price for selling this risk?

Simple Excel Certainty Equivalent Calc

		Utility = sqrt(x)	
Probability	Outcome	U(x)	
25%	100	10	
50%	200	14.14213562	
25%	500	22.36067977	
EV:	250		
EU:		15.16123776	
Certainty Equivalent:			
What is the c such that $EU = 100\% * u(c) = EU$ above?			
EU above:		15.16123776	
c:	229.8628		
u(c)	15.16123		
Difference:	-1.1E-05		

Goal Seek

Set cell:

To value:

By changing cell:

OK Cancel

Suppose outcomes are \$100 (25%), \$200 (50%), \$500 (25%). Utility function is $u(x) = \sqrt{x}$

$$EV = 25\% * 100 + 50\% * 200 + 25\% * 500 = 250$$

$$EU \text{ of risk} = 25\% * \sqrt{100} + 50\% * \sqrt{200} + 25\% * \sqrt{500} = 15.16123$$

Certainty equivalent: outcome \$c such that I get the same expected utility. So, $\sqrt{c} = 15.16123$
Or, $c = 229.86$

EU of risk MINUS EU of certainty equivalent

Set this difference to zero (so they are the same), by changing "c"

So, if I have $\sqrt{c}$ utility, I am indifferent between a gamble of \$100 (25%), \$200 (50%), and \$500 (25%), and a sure thing of \$229.86

Asymmetric Information

Seller has Private Information

- Asymmetric information leads to less efficiency than perfect information.
- If a buyer has uncertainty over a good's quality, he won't be willing to pay full price. This problem gets worse with risk aversion.
- If a seller knows more about the quality of a good than a buyer, they should signal that quality. However, lower-quality sellers will want to imitate.
 - So, the signal needs to be credible: it can't be worth it to low-quality sellers to imitate the signal. Example: warranties.
- The trick: find a signal that costs the "low quality" seller more.

Buyer has Private Information

- Buyers know more about their cost to serve (i.e. insurance)
- This can lead to adverse selection: only one type of person buys (i.e. high cost)
- Example: if you price insurance to average cost, you only attract individuals with above-average costs.
 - This can lead to a spiraling problem in insurance.
 - Instead, as with two-part tariffs, you want to screen and price to different pools.



Think about these problems in terms of supply and demand: will consumers be willing to buy at a certain price? Will sellers be willing to sell?

Moral Hazard

Unobserved Action

A **principal** hires an **agent** to carry out a task. Unfortunately, it is not possible (or, extremely costly) to monitor the agent's activities.

Basic principle: the agent will decide between options using EV (or EU if risk averse).
When designing an option (say, a bonus incentive), consider:

Do the Right Thing

The agent must get a higher payoff from taking the action that you want them to take.

Willing to Work

The agent must be willing to accept your offer over other alternatives.

Once you have designed an incentive scheme that satisfies this, you still need to make sure it is worth your while (i.e. paying the bonus doesn't offset the profit increase).

Efficiency Wage: Solve like an Infinitely Repeated Game

Instead of a bonus, you could pay more than your competitors and employees should work hard to avoid getting fired.

This is the end of MGT404....

- **But please stay in touch!**
 - **I love hearing about your current business challenges in economics, pricing, competition, etc**
 - **I love using examples provided by former students (it takes a village to teach economics...)**
 - **I am often available for the low price of an iced coffee :-)**
- **Please say hi**
- **Please complete the teaching evaluation.**

Good luck on the exam!

